Guojiao Zhao (Zoe)

(+86) 18315155220 zgj_1.17@sjtu.edu.cn
Shanghai Jiao Tong University Student Dormitory D3, Minhang, Shanghai, China 200240

EDUCATION

Shanghai Jiao Tong University *Sept. 2023 – Aug 2027*

- B.S. in Electrical and Computer Engineering, minor in Computer Science, Data Science. GPA: 3.76/4.0
- *Coursework:* Data Structures and Algorithms, Computer Vision, Intro to Signals and Systems, Probabilistic Methods in Eng., Design of Microprocessor Based Systems, etc.

Cornell University *Jan. 2026 - May 2026*

- Funded Exchange Program, GPA:3.89/4.0
- *Coursework:* AI Reasoning/Decision-Making, Intro to Machine Learning, Intro to Deep Learning, Optimization for AI.

RESEARCH EXPERIENCE

LLM-based Optimization Method *Research Assistant* *Feb. 2026 – Oct. 2026*

Advisor: Prof. Ruihao Zhu | Cornell University

- Developed an LLM-based optimization agent integrating mathematical formulation generation, solver execution, execution-log interpretation, and feedback-driven refinement for natural-language optimization problems.
- Built a supervisory agent to analyze recurring optimization modeling failures and generate reusable corrective instructions; evaluated agent performance using MIPLIB-NL and MIPLIB benchmarks.

Efficient and Precise Black-Box LLM Jailbreaking *Project Lead* *Jun. 2025 – Dec. 2025*

Advisor: Prof. David Woodruff | Carnegie Mellon University

- Proposed Semantic Steer Chain (SSC), a black-box LLM red-teaming framework based on progressive semantic approximation, and led the overall SSC design and implementation.
- Designed and conducted experiments on JailbreakBench dataset across 18 safety-aligned LLMs, achieving an 85.7% average attack success rate (ASR) and a 20+ percentage-point improvement over adaptive baselines.

Emotional Recognition based on Deep Learning *Research Assistant* *Jul. 2024 – Oct. 2025*

Advisor: Prof. Ruigang Yang | Shanghai Jiao Tong University

- Designed Morphological Convolutional Channel Local Attention module (MCCLA), a lightweight morphology-aware attention module to preserve subtle facial-expression cues and enhance local structural continuity.
- Improved RAF-DB accuracy from 90.68% to 91.23% with only 0.01M additional parameters.

Cold-start Algorithm in Recommender System *Funded Undergrad Researcher* *Mar. 2025 – Now*

Advisor: Prof. Xiaofeng Gao | Shanghai Jiao Tong University

- Investigated feature interference in user cold-start recommendation, focusing on how static user attributes and behavioral sequences contribute under sparse-history settings. Designed controlled feature-group ablation experiments and cold-ratio experiments.
- Explored multimodal representation learning for short-video recommendation by incorporating visual, audio, and textual features to support personalized recommendation.
- Investigated the integration of conventional recommendation algorithms with LLM-powered agents, enabling interactive responses to new users and personalized recommendation generation

PUBLICATION

Semantic Steer Chain: Efficient and Precise Black-Box Jailbreaking through Adaptive Approximation. The 30th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD). Hongkong, Jun. 2026. (Co-first author & Corresponding author)

LLMamba-Net: A Lightweight Network Integrating Linear Mamba for Facial Expression Recognition. The 8th Chinese Conference on Pattern Recognition and Computer Vision (PRCV). Shanghai, Oct. 2025.

AWARDS

- 2009 SEIEE Alumni Scholarship, Shanghai Jiao Tong University, 2024 (Top 20%)
- The Sam and Daisy Wu Scholarship [20000CNY], Global College, SJTU, 2025 (Top 10%)
- Interdisciplinary Contest in Modeling, Honorable Mention, 2025 (Top 20%)
- John Wu and Jane Sun Excellence Scholarship [50000CNY], Global College, SJTU, 2026 (Top 1%)

SKILLS

- **Languages:** Chinese (Native), English (TOEFL 5.5/6, R: 6/6, L: 5.5/6, S: 5/6, W: 5/6)
- **Technical Skills:** Python, C++, C, MATLAB, SQL, R
- **Certificates:** Level 3 in Shanghai College Information Technology Level Examination